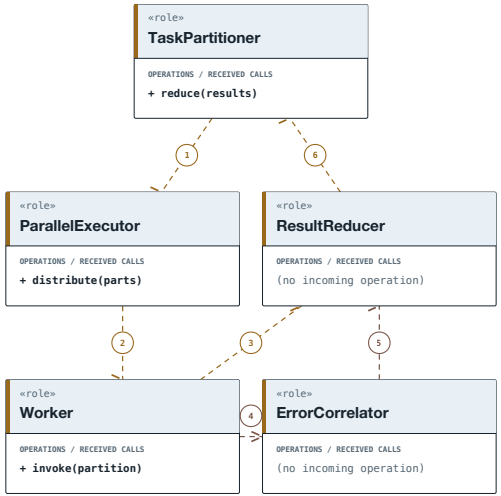
W03 Parallelization

Workflow / UML CLASS VIEW / ANALYTICAL MODEL

Run independent work or alternative judgments concurrently.

UML CLASS VIEW

dashed arrow: call dependency · brown: return/failure · analytical, not an observed trace



POINTCUT / PATTERN INTERFACE

CONTEXT	Independent tasks or alternative judgments can run concurrently.
TRIGGER	A coordinator fans work out and later aggregates it.
STRUCTURE	Partition contract, worker identity, result correlation, and reducer.
EFFECT	All selected results are reconciled into a declared aggregate or failure.

CONTRACT / REQUIRED BEHAVIOR

REQUIRES	Work partitions are independent enough for concurrency.
ENSURES	Aggregation includes every expected result or records its absence.
INVARIANT	Partial completion is never represented as full agreement.
VIOLATION	Failures, timeouts, and cancellation follow declared quorum or terminal policy.

RELATIONSHIPS / ANALYTICAL CALLS

number · caller → callee · message

1 TaskPartitioner → ParallelExecutor distribute(parts)
2 ParallelExecutor → Worker invoke(partition)
3 Worker → ResultReducer return(result)
4 Worker ↩ ErrorCorrelator report(error)
5 ErrorCorrelator ↩ ResultReducer correlate(errors)
6 ResultReducer → TaskPartitioner reduce(results)

VARIATION / TEST / NEGATIVE CONTROL

VARIATION	Sectioning, voting, map-reduce, and parallel tool calls.
TEST_OBLIGATION	Test disagreement, correlated failure, timeout, and one missing worker.
NEGATIVE_CONTROL	Bare Promise.all without a fan-in contract is only a cue.